New York University Tandon School of Engineering
Mechanical and Aerospace Engineering

ROB-GY 6013 Mathematics for Robotics

Problem Set 3: Dynamics and Probability

Due Wednesday, November 18, 2026, before class (5:00 PM)
Submit on Gradescope.

Discussion with other students is allowed. Write your solutions alone.
Use any tool to understand the material, but not to write your solutions.
Solutions typeset in LaTeX or Markdown earn a 10% bonus. Show your reasoning: a bare final answer earns little credit.

Problem 1: Linear ODEs, Euler stability, and Runge–Kutta (20 points)

A robot joint with a spring and a damper obeys

$$\ddot{q} + 5\dot{q} + 4q = 0, \qquad q(0) = 1, \quad \dot{q}(0) = 0.$$

A simulator does not solve this equation exactly: it advances the state $x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$ in steps of size h. In this problem you compare the exact solution with two such methods, Euler and RK4.

Parts (c) and (d) use the starter notebook `ps3-ode.ipynb`. You do not submit the notebook: copy the code you wrote in the TODO cell into your write-up, and answer the questions below there.

(a) (6 pts) Write the equation as $\dot{x} = Ax$. Find the eigenvalues and eigenvectors of A, and use them to find $q(t)$.

(b) (4 pts) Euler's method gives $x_{k+1} = x_k + hAx_k = (I + hA)\, x_k$. Find the eigenvalues of $I + hA$, and the range of step sizes $h > 0$ for which the Euler states decay to 0 from every starting state.

(c) (6 pts) In the notebook, complete one step of Euler and one step of RK4, then run the error table and include it. By about what factor does the error of each method shrink when h is halved, and what does this say about the order of each method? How many evaluations of $f(x) = Ax$ do Euler with $h = 0.0125$ and RK4 with $h = 0.05$ use to reach $t = 2$, and which is more accurate?

(d) (4 pts) Run the last cell, which uses $h = 0.625$, and include its plot. What happens to the Euler solution? Explain it using (b). Does RK4 decay at this step size?

Problem 2: Gaussian conditioning and the Kalman gain (20 points)

A mobile robot estimates its distance x to a wall, in meters. Before any reading it believes $x \sim \mathcal{N}(\mu_0, P_0) = \mathcal{N}(10, 2)$. A lidar then reads $z_1 = 11$ with noise variance $R_1 = 1$, and a sonar reads $z_2 = 13$ with noise variance $R_2 = 2$. Both sensors measure the same wall distance x. For sensor $i \in \{1, 2\}$, the reading is $z_i = x + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, R_i)$. Assume $x, \varepsilon_1, \varepsilon_2$ are mutually independent. Combining a prediction with a reading in this way is the measurement update of a Kalman filter.

(a) (5 pts) Update the belief with the lidar reading: compute the gain K, the posterior mean μ_1 and the posterior variance P_1.

(b) (5 pts) Use the result of (a) as the prior and update with the sonar reading. Compute the gain, the posterior mean μ_2 and the posterior variance P_2.

(c) (5 pts) Show that one update with prior $\mathcal{N}(\mu, P)$ and a reading z with noise variance R gives

$$\frac{1}{P'} = \frac{1}{P} + \frac{1}{R}, \qquad \mu' = P'\Big(\frac{\mu}{P} + \frac{z}{R}\Big).$$

Use it to check (b). Does the order of the two readings matter?

(d) (5 pts) Suppose the robot has no prior information, $P_0 \to \infty$. What are the mean and variance of the fused estimate? Which reading gets more weight, and why?

Problem 3: Maximum likelihood, MAP, and regularized least squares (20 points)

In Problem Set 2, Problem 1, you fitted a linear model to the data below by least squares and then added a penalty that discourages large weights. This problem shows that both come from Gaussian assumptions. The model is $b = A\theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, and the data are

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}.$$

(a) (5 pts) Write the likelihood $p(b \mid \theta)$. Show that maximizing it over θ is the same as minimizing $\frac{1}{2}\|b - A\theta\|^2$, and give the maximizer $\hat{\theta}_{\mathrm{MLE}}$ for the data.

(b) (5 pts) Now add the prior $\theta \sim \mathcal{N}(0, \tau^2 I)$. Show that maximizing the posterior $p(\theta \mid b)$ is the same as minimizing

$$J(\theta) = \frac{1}{2}\|b - A\theta\|^2 + \frac{\lambda^2}{2}\|\theta\|^2, \qquad \lambda^2 = \sigma^2/\tau^2.$$

(c) (5 pts) Solve $\nabla J(\theta) = 0$ for $\hat{\theta}_{\mathrm{MAP}}$, and compute it for $\sigma^2 = \tau^2 = 1$. Compare it with $\hat{\theta}_{\mathrm{MLE}}$ from (a).

(d) (5 pts) What does $\hat{\theta}_{\mathrm{MAP}}$ tend to as $\tau^2 \to \infty$, and as $\tau^2 \to 0$? Explain each limit in terms of how much the prior is trusted.

Problem 4: Entropy, cross entropy, and KL divergence (20 points)

A robot logs the outcome of every grasp as success (s), slip (l) or miss (m). Over many grasps the outcomes occur with probabilities $p = (p_s, p_l, p_m) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. The log stores each outcome with a binary prefix code: no codeword is the beginning of another. All logarithms are base 2.

(a) (4 pts) Compute the entropy $H(p)$. Give a prefix code with codeword lengths $\ell_i = -\log_2 p_i$, and check that its average length equals $H(p)$.

(b) (5 pts) An engineer believes the probabilities are $q = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$. Compute the cross entropy $H(p, q)$ and $D_{\mathrm{KL}}(p \,\|\, q)$. Based on q, the engineer chose codeword lengths $\ell_i = -\log_2 q_i$. Compute the average code length when the outcomes actually follow p, and check that it equals $H(p, q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$.

(c) (5 pts) A second model $r = (\frac{1}{2}, \frac{1}{2}, 0)$ never predicts a miss. Compute $D_{\mathrm{KL}}(p \,\|\, r)$ and $D_{\mathrm{KL}}(r \,\|\, p)$. Is D_{KL} symmetric, and can it be used as a distance between distributions?

(d) (2 pts) A model that outputs probabilities $q = (q_s, q_l, q_m)$ is trained on N independent logged grasps by maximizing the likelihood of their outcomes $o_1, \ldots, o_N \in \{s, l, m\}$. Show that this is the same as minimizing the loss $-\frac{1}{N} \sum_{n=1}^{N} \log_2 q(o_n)$, where $q(o_n)$ is the probability that q assigns to outcome o_n.

(e) (4 pts) Show that this loss approaches $H(p, q)$ as N grows. Then use $H(p, q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$ to explain which q minimizes $H(p, q)$, and which term training can remove.